RELATIVE PERFECTION DESCENDS UPWARD THROUGH A FINITE ORDER THICKENING OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13138-13151 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 13153-13168, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source definition-relatively-perfect, lemma-thickening-pseudo-coherent, lemma-affine-locally-rel-perfect; Derived Categories of Spaces, lemma pseudo-coherent; More on Morphisms, lemma thickening-relatively-perfect.

Source reference. Original lemma 0DKN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12271–12282. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12313–12335. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DJX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13081–13085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DJY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13168–13168. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08JL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13168–13168. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Consider a cartesian diagram*

$$\begin{array}{ccc} X & \xrightarrow{i} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xrightarrow{j} & Y' \end{array}$$

of algebraic spaces over S. Assume $X' \to Y'$ is flat and locally of finite presentation and $Y \to Y'$ is a finite order thickening. Let $E' \in D(\mathcal{O}_{X'})$. If $E = Li^*(E')$ is Y-perfect, then E' is Y'-perfect.

Proof. Recall that being Y-perfect for E means E is pseudo-coherent and locally has finite tor dimension as a complex of $f^{-1}\mathcal{O}_Y$-modules (Definition 0DKN). By Lemma 0DJX we find that E' is pseudo-coherent. In particular, E' is in $D_{QCoh}(\mathcal{O}_{X'})$, see Derived Categories of Spaces, Lemma 08JL. By Lemma 0DKQ this reduces us to the case of schemes. The case of schemes is More on Morphisms, Lemma 0DJY. □